Silicon Photonic MZIs and Bragg Filters: Design, Process Variation and Thermal Analysis of the v11 Circuit Set

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Abstract-- A five-circuit silicon photonic design integrates three Mach-Zehnder interferometers (MZIs) and two Bragg filters within a 605 × 410 µm layout. The updated evidence comprises 714 native circuit cases, 1,432 complex spectra and 6,308,000 complex samples. Nominal local free spectral ranges at 1550 nm are 22.93 nm for A25, 5.7326 nm for A100, and 5.8386/5.8360 nm for the two Course500 outputs. Doubling the Bragg period count from 150 to 300 lowers external transmission at 1550 nm by 29.48 dB. Paired 100-trial process ensembles quantify conditional geometry sensitivity. A separate fitted-material MODE study gives a group index of 4.21154 and a first-order +10 K fringe-shift estimate of +0.753 nm. The tested native configurations showed no spectral change with temperature. Mesh sensitivity, interpolation accuracy and truncated Bragg bands limit the scope of the conclusions.

Index Terms-- Silicon photonics, Mach-Zehnder interferometer, Bragg grating, process variation, Monte Carlo, thermo-optic sensitivity.

I. Introduction

Unequal-arm MZIs connect waveguide phase dispersion to free spectral range (FSR), while finite Bragg filters provide wavelength-selective suppression [1]. This design compares path imbalance and grating period count on one independently addressable layout.

The v11 layout was established on 5 October 2026. The 11 October simulations add updated MODE dispersion and INTERCONNECT process-variation results, including both Course500 outputs and all three Bragg channels. These calculations form the simulation study presented here; the layout geometry is unchanged.

The results describe simulations under explicitly selected material, circuit and process assumptions. They do not represent fabricated-device measurements or production yield. Source identities, numerical audits and coverage gaps are stated in this report.

The full study retains the preliminary 600 nm MODE and ideal-MZI analysis, the 5 October 500 nm baseline, and the 11 October extended simulations. These stages use distinct models and extraction methods; earlier results are retained explicitly rather than substituted for the updated results.

II. Theory And Evidence Scope

For an ideal balanced-amplitude MZI with path difference ΔL, the phase and normalized power are

$$\beta = 2\pi n_{eff}/\lambda, \quad T = [1 + \cos(\beta \Delta L)]/2. \tag{1}$$

The group index and local propagation-only FSR are

$$n_g = n_{eff} - \lambda \, dn_{eff}/d\lambda, \tag{2}$$

$$FSR_\lambda \approx \lambda^2/(n_g \Delta L). \tag{3}$$

Complete circuit spectra also contain coupler phase, grating-coupler envelopes, routing loss and interference from secondary paths. Equation (3) is therefore a comparison scale, not a replacement for full-circuit extraction.

TABLE I
SAVED NATIVE CIRCUIT COVERAGE

Circuit	Cases	Ports	Points / trace
A25	142	1	5,000
A100	142	1	5,000
Course500	142	2	2,000
BraggN150	144	3	5,000
BraggN300	144	3	5,000

Each circuit shares 42 deterministic and 100 Monte Carlo (MC) geometry cases at 300 K. Each Bragg batch additionally contains nominal 305 K and 310 K diagnostics. These totals exclude earlier trial runs and separate MODE solves.

MODE solves uniform waveguide cross sections; INTERCONNECT uses the native EBeam compact models [2]. Their nominal indices differ. In particular, the fitted 400-cell MODE result is n_{eff} = 2.445721947 and n_g = 4.211238804, whereas the native nominal lookup gives 2.445530000 and 4.190880000. These baselines are kept separate throughout.

III. Design Rationale And Analytical Method

A. Controlled comparisons and design scope

The v11 circuit set supports two controlled comparisons: changing MZI path imbalance and changing Bragg length. The rationale below explains what the selected geometry tests; it does not claim an undocumented optimization or a pre-established numerical acceptance specification.

The 500 × 220 nm waveguide defines a common nominal cross-section for the layout, native lookup model and updated MODE study. A shared cross-section makes model differences and geometry sensitivity interpretable without also changing the nominal waveguide. The earlier 600 nm exercise is a separate baseline. The available evidence does not establish that 500 nm is globally optimal or single-mode across every geometry and wavelength.

B. Why compare 25 and 100 µm?

A25 and A100 use the same nominal waveguide and a fourfold difference in path imbalance. With equal group index and negligible coupler dispersion, FSR scales inversely with ΔL. Thus A100 should have approximately one quarter of A25's FSR. The simulated values, about 5.73 and 22.93 nm, test this scaling directly.

A larger imbalance increases phase change per unit frequency and provides denser fringes; it also reduces the unambiguous interval between fringe orders and can increase differential attenuation. A25 offers wider fringe spacing. Neither choice is assigned a sensing-resolution or yield target that was not specified before this analysis.

C. Why retain Course500 and the layout constraints?

Course500 denotes the inherited course design with a 500 nm nominal waveguide, not a 500 µm arm imbalance. Its approximately 99.294 µm imbalance is close to A100, while its directional-coupler combiner and two outputs differ from the Y-combiner circuits. It is therefore a useful topology comparison at nearly equal propagation imbalance.

The 127 µm grating-coupler pitch organizes fibre access; fixed landing locations, the 605 × 410 µm boundary and 5 µm routing bends support the layout implementation. These are layout constraints and inherited component choices, not evidence of an independently optimized bend-loss or fibre-coupling design.

D. How imbalance produces the spectrum

Let a_1 and a_2 be real, nonnegative arm amplitudes at a selected output, including splitting, combining and attenuation. Their phase difference is Δφ. Coherent addition gives

$$P = a_1^2 + a_2^2 + 2a_1a_2 \cos \Delta\varphi. \quad (4)$$

For balanced arms, normalization gives $(1 + \cos \Delta\varphi)/2$; the other output can have a complementary phase. Unequal amplitudes leave a minimum $(a_1 - a_2)^2$, explaining why phase alignment alone does not ensure infinite extinction.

$$\Delta\varphi = \beta\Delta L + \theta, \beta = \omega n_{eff}/c. \quad (5)$$

Here θ includes the net component phase. Differentiating at fixed geometry uses $n_g = n_{eff} + \omega\, dn_{eff}/d\omega$ and yields the differential delay

$$\Delta\tau = d\Delta\varphi/d\omega = n_g\Delta L/c + d\theta/d\omega. \quad (6)$$

Adjacent fringes differ by 2π. Over a locally smooth band,

$$\Delta f \approx 1/|\Delta\tau|, FSR_\lambda \approx \lambda^2/(c|\Delta\tau|). \quad (7)$$

Equation (3) follows when component-phase dispersion is neglected. The full circuit includes those phases and wavelength-dependent envelopes. Course500's approximately 1.12% FSR difference from the propagation estimate may involve component dispersion, extraction and model differences; these contributions have not been isolated.

E. How the thermal estimate is obtained

Following the same fringe order at fixed ΔL and fixed component phase requires $\delta(\Delta\varphi) = 0$. Linearizing the propagation phase gives

$$0 \approx -(2\pi n_g\Delta L/\lambda^2)\delta\lambda + (2\pi\Delta L/\lambda)\delta n_{eff}. \quad (8)$$

$$\delta\lambda \approx \lambda\, \delta n_{eff}/n_g. \quad (9)$$

The imbalance cancels at first order: increasing ΔL increases phase sensitivity but does not by itself multiply the same-order thermal wavelength shift. The estimate excludes thermal expansion, changing coupler phase and differential arm heating. It is evaluated from MODE index changes and is not a temperature-calibrated native circuit simulation.

IV. Preliminary 600 Nm Mode Study

The first waveguide study uses a 600 × 220 nm silicon strip and the fundamental quasi-TE solution at 1550 nm. It is retained as the starting point for the ideal-MZI exercise; it is not the 500 nm cross-section used by the native v11 circuits.

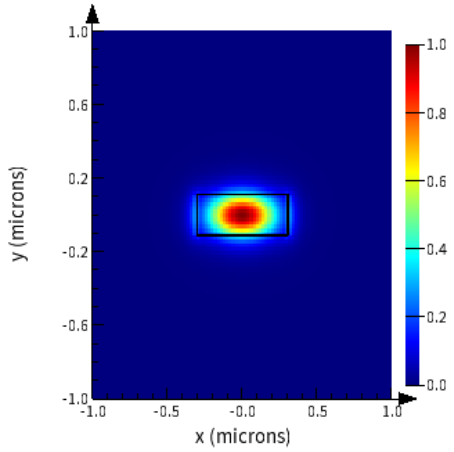


Fig. 1. Original normalized electric-field intensity at 1550 nm for the preliminary 600 × 220 nm quasi-TE mode. This is the modal field figure retained from the initial study.

The field is concentrated in the silicon core. At 1550 nm, n_{eff} = 2.571982, n_g = 4.104976 and the displayed TE polarization fraction is 99%. The ideal material calculation reports zero imaginary effective index and zero simulated propagation loss; these are not measured fabrication losses.

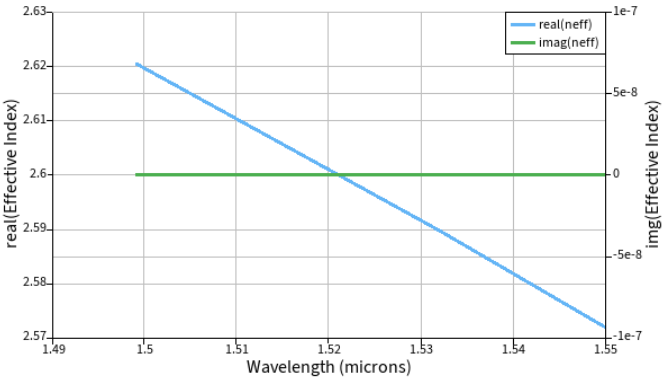


Fig. 2. Original 600 nm effective-index display over 1500-1550 nm. The numerical export spans 1500-1600 nm with 101 points; this retained display shows only its shorter-wavelength portion.

A. Dispersion and local compact-model fit

The real effective index decreases with wavelength. The exported group-index curve contains a discontinuity near 1532 nm while tracked-mode overlap remains close to unity. It was treated as a suspected sweep/model-data artifact, not demonstrated to be a physical transition; its precise cause remains unresolved.

The preliminary compact model therefore fits only the smooth 1540-1560 nm interval. Restricting the fit avoids using the discontinuity in the local MZI model, but does not validate the excluded part of the sweep.

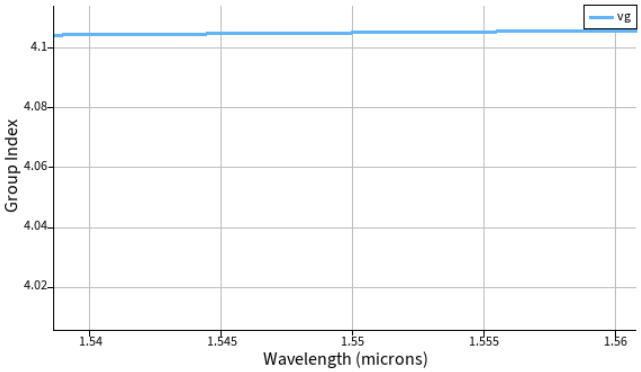


Fig. 3. Original 600 nm group index over the retained 1540-1560 nm compact-model window.

With u = λ - 1.55 and λ expressed numerically in micrometres, the retained cubic is

$$n_{eff}(\lambda) = 2.5719816674 - 0.9890283430u - 0.0220137755u^2 + 0.0091898081u^3. \quad (10)$$

Its maximum absolute residual in the fit window is 9.3 × 10^{-11}. Differentiation using (2) gives n_g(1550 nm) = 2.5719816674 + 1.55 × 0.9890283430 = 4.1049755991, matching the rounded direct MODE value.

This small residual measures the agreement of a polynomial with the supplied numerical samples. It does not establish mesh convergence, material accuracy or physical waveguide loss. The preliminary polynomial is not used to replace the native EBeam lookup in the later 500 nm circuits.

B. Why retain this preliminary study?

The modal field shows which polarization and confinement pattern were selected. The dispersion fit connects the field solution to a circuit-level phase model. Together they document the MODE-to-MZI reasoning that precedes the later native-model comparison.

V. Preliminary Ideal-MZI Simulation

The preliminary INTERCONNECT circuits use an ideal lossless 50:50 Y splitter, two waveguides using the 600 nm MODE model, and an ideal Y combiner. The upper arm is 100 µm; lower arms are 125, 150, 175, 200 and 225 µm. The ONA sweeps 1500-1550 nm with 2001 samples, giving 25 pm spacing.

TABLE II
PRELIMINARY 600 NM IDEAL-MZI RESULTS AT 1525 NM

ΔL (µm)	Sim. FSR (nm)	Theory FSR (nm)	Extracted n_g
25	23.070	22.662	4.032
50	11.436	11.331	4.067
75	7.652	7.554	4.052
100	5.730	5.665	4.059
125	4.582	4.532	4.061

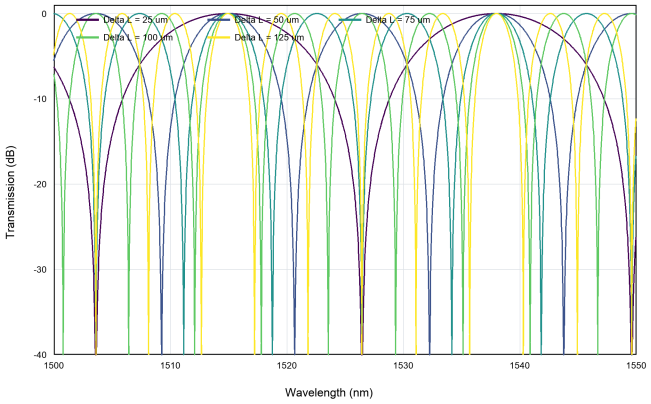


Fig. 4. Original ideal-MZI transmission spectra for five path imbalances in the preliminary 600 nm model.

Increasing path imbalance produces denser interference fringes, as expected from (3). The table retains the original mean-FSR extraction at the nominal 1525 nm comparison wavelength. It is distinct from the later local-frequency interpolation at 1550 nm.

The 1540-1560 nm polynomial fit window does not cover the entire 1500-1550 nm ONA sweep. Results below 1540 nm should therefore be regarded as a preliminary circuit exercise outside the demonstrated local fit window, not broadband validation of that fit.

A. Inverse-imbalance fit and extracted group index

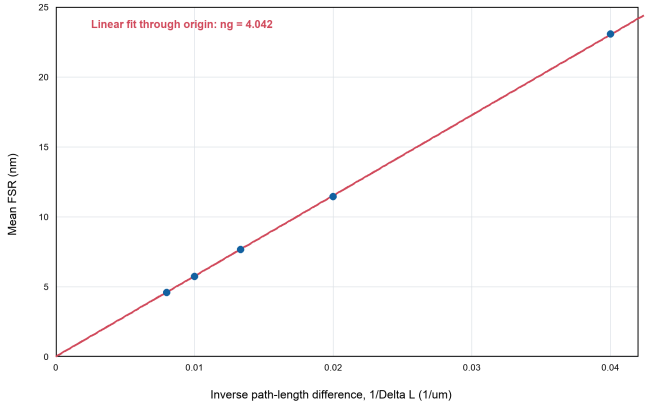


Fig. 5. Original mean FSR versus inverse path imbalance. The origin-constrained fit gives $n_g = 4.042$ for the preliminary circuit exercise.

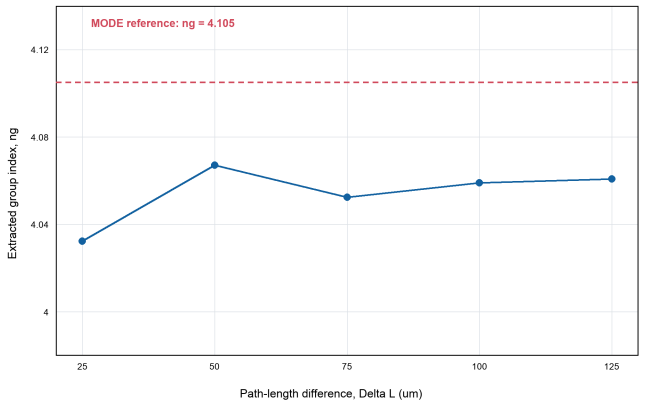


Fig. 6. Group-index estimates extracted separately from the five original ideal-MZI spectra.

Writing FSR = m/ΔL gives $n_g \approx \lambda_0^2/m$. The combined estimate 4.042 differs from the direct 600 nm MODE value 4.104976 by approximately -1.54%. This is a model/extraction comparison, not a measured index error.

The 25 µm case contains only about two complete fringe periods within the sweep, so a mean spacing is less statistically robust. Larger imbalances provide more extrema. Finite sampling, dispersion across the averaging band and use outside the local fit window can affect the comparison; their individual contributions were not isolated.

This exercise establishes the qualitative 1/ΔL trend and motivates testing several imbalances. Its five ideal circuits and 600 nm waveguide must not be pooled with the three native 500 nm MZI datasets or the later paired MC100 trials.

VI. Original 500 Nm Mode Baseline

The separate 5 October study uses a 500 × 220 nm Si (Silicon) - Palik core, a uniform fixed-index background of 1.44401, and a 2D Z-normal FDE solver at 300 K. Its 2 × 2 µm domain has 100 × 100 cells and Metal boundaries on all four transverse sides. Screenshots documented these settings; the corresponding LMS project was not supplied at that stage.

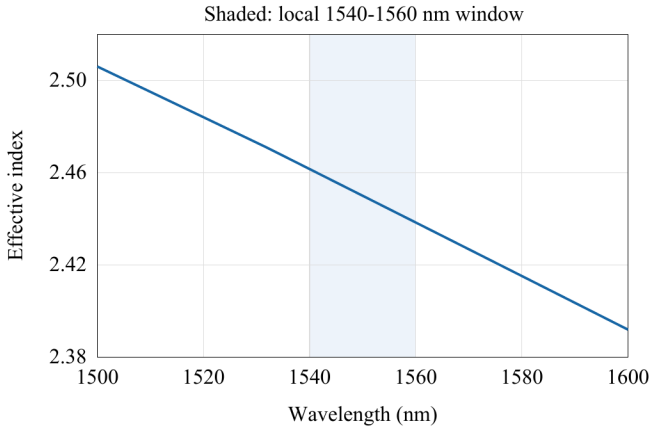


Fig. 7. Complete original 101-point 500 nm effective-index export over 1500-1600 nm. Shading marks the local fitting interval.

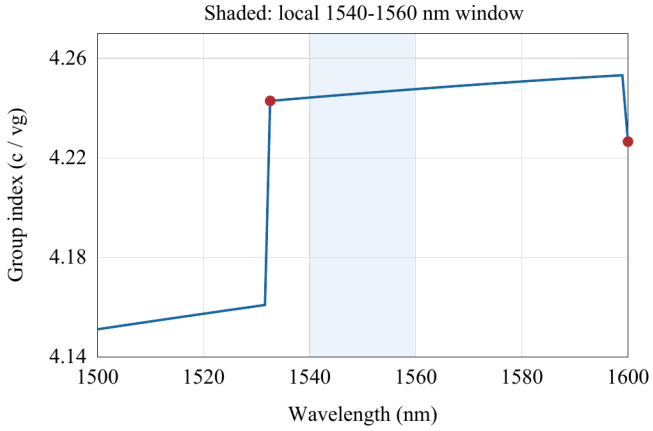


Fig. 8. Original 500 nm group index computed as c/v_g. Red markers retain the step near 1532.6 nm and endpoint anomaly at 1600 nm.

Mode tracking and detailed dispersion were enabled. Overlap spans 0.9999962-0.9999997; the 1550 nm screen reports 98% TE polarization. Exported imaginary index and loss are zero under the original settings.

A. Local fit and comparison with the original circuits

Linear interpolation at 1550 nm gives $n_{eff} = 2.450006$ and $n_g = 4.245970$. A cubic fitted to the 20 samples in 1540-1560 nm gives the same rounded group index through (2), agreeing with c/v_g to better than 10^{-6}. The maximum effective-index residual is 2.49×10^{-10}.

TABLE III
ORIGINAL 500 NM GROUP-INDEX COMPARISON AT 1550 NM

Method / model	n_g	Difference (%)
100-cell MODE, c/v_g	4.245970	Reference
100-cell MODE, local fit	4.245970	<0.0001
Original A25 extraction	4.190280	-1.3116
Original A100 extraction	4.189874	-1.3212

Differences use $100(n_{g,circuit}/n_{g,MODE} - 1)$. These are the original sampled-minimum estimates, not the updated extraction values later in this report.

The export rises from 4.160908 at 1531.589 nm to 4.242887 at 1532.567 nm. At its endpoint, it falls from 4.253211 at 1598.934 nm to 4.226535 at 1600 nm. Neither feature lies within the local fit window. Near-unity overlap supports mode continuity but does not establish the anomaly mechanism.

B. Relationship to the 11 October update

No mesh/domain convergence study accompanied this original baseline. The later fitted-material study adds 400/800-cell and domain comparisons, with their own documented limitations. Its nominal n_g is approximately 4.21154 rather than 4.24597. Because material fitting and discretization both change, the difference is not assigned solely to mesh refinement.

The original complete 1500-1600 nm curves remain visible here. The newer dense nominal export covers only 1500-1550 nm; it does not erase or fill the original longer-wavelength record. The native lookup $n_g = 4.19088$ is a third model baseline. A close numerical comparison does not establish equivalent material stacks or dispersion.

VII. Original Native-circuit Baselines

A. A25/A100 sampled-minimum analysis

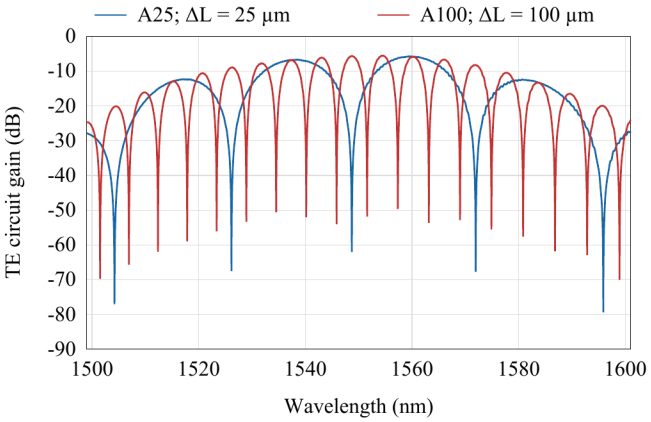


Fig. 9. Original A25/A100 ONA input 1, mode 1 TE gains, each containing 5,000 samples over 1499.019-1601.000 nm.

TABLE IV
ORIGINAL NATIVE MZI EXTRACTION

Quantity	A25	A100
Routed arms (µm)	363.975 / 388.975	363.975 / 463.975
ΔL (µm)	25.000	100.000
FSR at 1550 nm (nm)	22.934	5.734
Inferred n_g	4.1903	4.1899
Minima / intervals	5 / 4	18 / 17

The listed arms exclude two matched 20 µm straight sections per branch; including them leaves ΔL unchanged. Gain and wavelength rows were paired before sorting. Strict sampled minima below -35 dB defined adjacent intervals without smoothing or notch interpolation.

For ordered adjacent minima $\lambda_1 < \lambda_2$, the original interval estimate was

$$n_{g,int} = \lambda_1\lambda_2/[\Delta L(\lambda_2 - \lambda_1)]. \quad (11)$$

Interval indices were interpolated between midpoint wavelengths to 1550 nm, then converted to FSR using (3). The original FSR ratio was 3.9996. The later method refines principal minima and interpolates frequency intervals, explaining the small numerical update.

The A100 file retained the header a25_wavelength_nm; its numerical grid matched A25. Original circuit identity relied on supplied filenames because ICP projects were unavailable for that evidence set.

B. Original Bragg period-count comparison

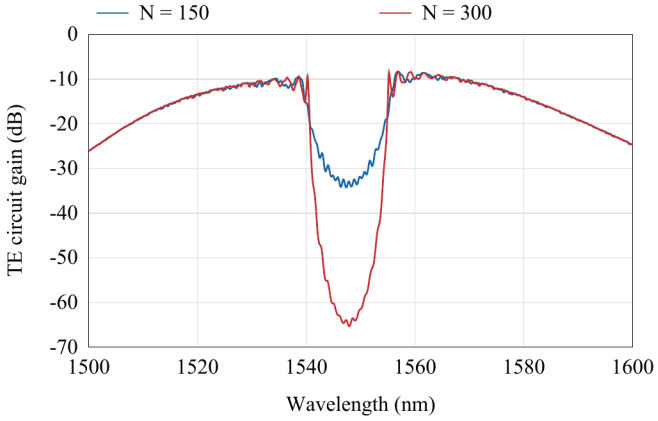


Fig. 10. Original Bragg ONA input 3, mode 1 transmission gains: 2,000 samples over 1500.047-1600.000 nm.

TABLE V
ORIGINAL BRAGG EXTERNAL TRANSMISSION

Quantity	N150	N300
Length (µm)	47.55	95.10
Sampled minimum (dB)	-34.242	-65.310
Minimum wavelength (nm)	1547.339	1547.888
Gain at 1550 nm (dB)	-31.996	-61.477

These exports originated from v10. The v11 implementation retained the Bragg silicon, pins and device-recognition geometry, allowing the same spectra to be retained as a labelled earlier baseline. They are not the newly rerun three-output 11 October batch.

Linear interpolation in dB at 1550 nm gave N300 29.480 dB lower transmission. Grating couplers, the directional coupler and routing contribute to this external response. Different sampled-minimum wavelengths do not prove a physical centre shift.

C. What the new data resolve

Numerical reference-channel spectra were unavailable in the original set, so no normalized suppression or bandwidth was claimed then. The 11 October runs provide reflection, reference and transmission together, enabling the explicitly defined ratios and half-power bands later in this report.

The native compact model still does not separately characterize the 10 nm terminal landing used to meet the 500 nm declared port width. New spectra improve numerical coverage but do not independently validate that geometry or the grating physics.

VIII. Original Course500 Evidence And Study Progression

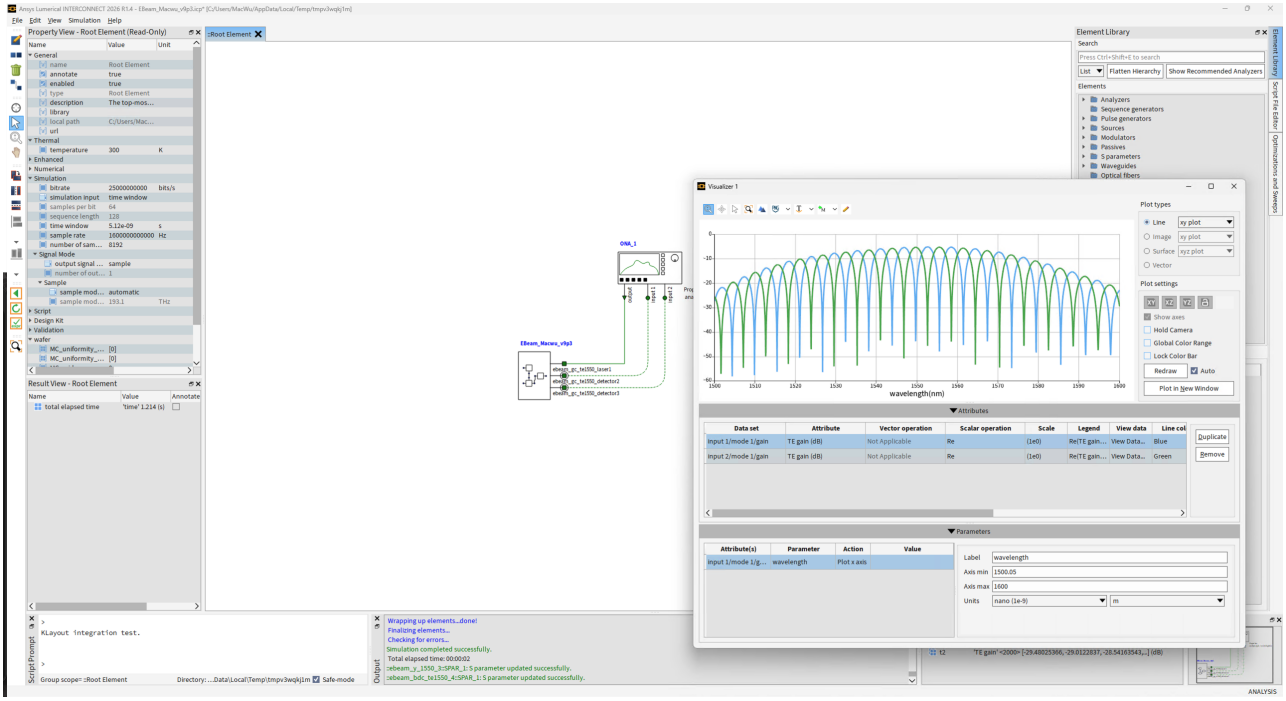


Fig. 11. Original Course500 screen supplied on 4 October 2026. The displayed project is EBeam_Macwu_v9p3, with a completed run and alternating output spectra. It is retained as historical qualitative evidence; numerical v11 results are presented separately.

A. Inherited circuit and current implementation

The course exercise uses a TE grating input, Y splitter, unequal 500 nm waveguide arms, a broadband directional-coupler combiner and two grating outputs. The original layout ledger gives $\Delta L = 99.293$ µm; the rounded current netlist gives 99.294 µm.

The early screenshot supports qualitative two-output behavior only. At that stage paired numerical outputs were unavailable, so no numerical FSR was inferred from the image. The 11 October dataset now supplies both outputs, nominal spectra, corners and paired MC100 trials; the former missing-data statement is historical, not a current limitation.

The v11 layout retains the Course500 and Bragg geometry from v10. For A25/A100, four former taper sections per circuit became 20 µm straight 500 nm sections. The preliminary 600 nm compact model is not inserted into any of these native v11 circuits.

B. Measurement plan and interpretation

A future optical measurement should excite row F2 in TE at each circuit column and collect the designated output rows: F3 for A25/A100, F3/F4 for Course500, and F1/F3/F4 for Bragg reflection/reference/transmission. Acquiring all Bragg channels is necessary to apply the stated reference normalization.

The report now preserves the progression from preliminary modal confinement and ideal interference, through the first 500 nm baseline, to extended native-circuit variation. These are simulations, not fabricated measurements. Local connectivity/rule checks remain distinct from foundry signoff.

The following sections retain the current v11 layout and the full 11 October analysis. Their finer MODE mesh, explicit material fitting, updated spectral extraction and larger circuit ensembles are stated directly so that improvements and unresolved limits can be assessed directly.

IX. Layout And Circuit Implementation

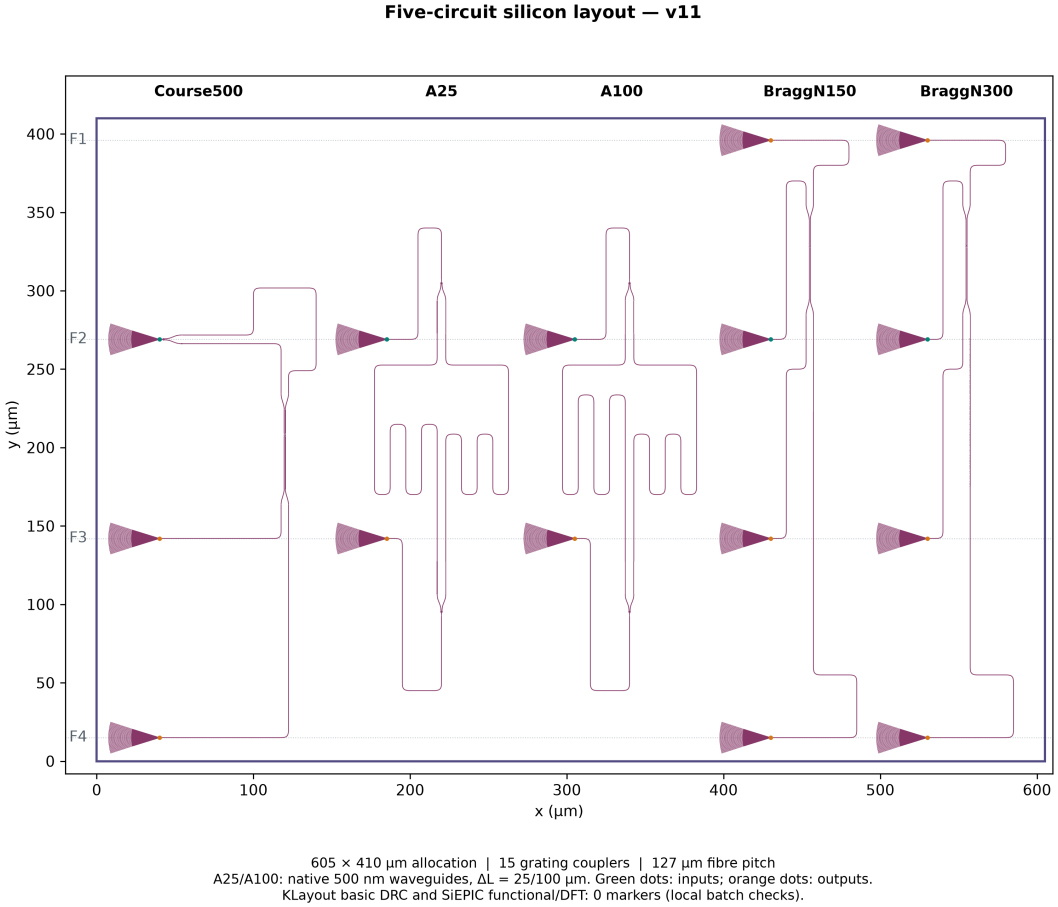


Fig. 12. Preserved EBeam_Macwu_v11 silicon layout. Green and orange markers identify input and output grating-coupler pins. This is the same v11 geometry used as the layout baseline; the October update adds simulation evidence.

The five circuits occupy 605 × 410 µm. Fifteen grating couplers use 127 µm vertical fibre pitch and R0 orientation. F2 at y = 269 µm is the common input row. A25/A100 use F3 outputs; Course500 uses F3 and F4. Each Bragg uses F1 for reflection, F3 for reference and F4 for transmission.

Pin columns lie at x = 40, 185, 305, 430 and 530 µm. Fibre targets are 20.4 µm left of the pins. Connections use nominal 500 nm waveguides and a 5 µm TE bend setting.

A25 arms are 363.975/388.975 µm and A100 arms are 363.975/463.975 µm, excluding two matched 20 µm straight sections in each branch. Their imbalances are 25 and 100 µm. Course500 retains the Y-splitter and broadband directional-coupler combiner; its current rounded netlist imbalance is 99.294 µm (the layout ledger reports 99.293 µm).

Both rectangular Bragg profiles have 317 nm period and a 50 nm corrugation parameter, corresponding to alternating full widths of 450 and 550 nm about a 500 nm bus. Zero sidewall misalignment is specified. N = 150 and 300 give grating lengths of 47.55 and 95.10 µm.

Layout verification for this design produced 136 deterministic checks, zero basic and functional layout markers, 55 components, 53 nets and 106 connected optical pins. Five unique opt_in labels support separate circuit access. Recorded tools are KLayout 0.30.12, SiEPIC Tools 0.5.31 and EBeam PDK 0.4.53.

These checks assess layout connectivity and the stated geometric rules. They do not constitute foundry signoff or a manufacturing certificate. The subsequent simulation updates do not alter the v11 geometry.

X. Waveguide Model And Dispersion

A. Fitted-material configuration

The updated 500 × 220 nm FDE study uses a copied Si (Silicon) - Palik material, a uniform background index of 1.44401 at 300 K, and a 2D Z-normal solver with Metal transverse boundaries. The usual domain is 2 × 2 µm with 400 × 400 cells; the refined nominal sweep uses 800 × 800 cells.

TABLE VI
MODE MATERIAL-FIT SETTINGS

Setting	Adopted value
Simulation fit band	1.5-1.6 µm
Tolerance	0.0001
Coefficient limit / actual	20 / 2
Model flags	MCM 1; sampled 1; analytic 0
Fit options	Improve stability; make passive
Batch trial modes	3
Standard / weighted RMS	0.0013975 / 0.0013975

A preliminary tolerance of 0.1 produced zero coefficients and a nearly flat fitted index. That setting was rejected for dispersion analysis. The retained material-fit RMS is a fit residual; it is not a bound on modal-index or device-wavelength error.

Both recorded RMS values are 13.975 times the requested 0.0001 tolerance: the fit does not meet its target. The solver may return its best available fit when that target is unattainable [3]. Raising the coefficient cap from 10 to 20 retained two coefficients and the same residual. Disabling stability improved RMS only to 0.00139578 while increasing the fitted imaginary-index scale from roughly 10^{-9} to 10^{-6}.

The stable fit is retained provisionally for dispersion comparison, not as an accurate material calibration. The band follows simulation settings; the explicit fit-range override was unchecked. Exact fitted-project persistence remains incomplete. Group-index consistency cannot validate material accuracy.

B. Mesh and domain sensitivity

At six paired geometries and temperatures, refinement from 400 to 800 cells changes n_{eff} by 8.4421×10^{-5} to 9.9252×10^{-5} and n_g by 2.5442×10^{-4} to 3.3945×10^{-4}. The proposed 10^{-5} effective-index target is not met; the 10^{-3} group-index target is met by these differences. Full values appear in Table XVII.

At nominal geometry and 300 K, the refined-minus-coarse phase-equivalent wavelength scale is 0.03319 nm. This estimates sensitivity between two discretizations, not error relative to a continuum solution.

Doubling the domain from 2 to 4 µm at a nominal 5 nm step changes n_{eff} by -1.03226×10^{-7} and n_g by $+2.58869 \times 10^{-6}$. The trial-mode count also changes from 20 to 3, so the domain comparison retains that confounder.

C. Nominal wavelength sweep

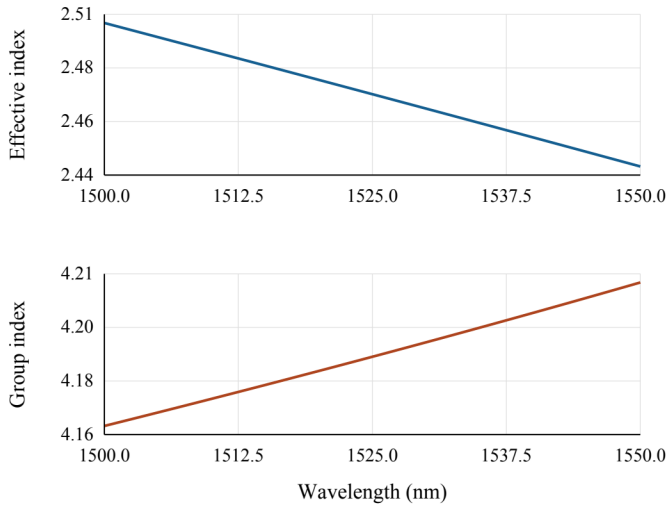


Fig. 13. Updated fitted-material nominal sweep, 800 cells per transverse axis, 300 K. The actual 101-row export covers 1500-1550 nm only; no dense values above 1550 nm are inferred.

At 1550 nm, the export gives $n_{eff} = 2.445812142538$ and $n_g = 4.211540509692$. Across the received band, n_g ranges from 4.16117678 to 4.21154051. Tracking overlap ranges from 0.99999812 to 0.99999919, supporting continuity of the selected mode within this sweep.

A finite-difference evaluation of (2) agrees with c/v_g to a maximum absolute difference of 1.04695×10^{-6}. This is an internal consistency check within the adopted model. It does not establish material accuracy or validate unreceived wavelengths.

The 1550 nm fitted loss is 0.0535823 dB/m (0.000535823 dB/cm). It arises from the small fitted imaginary index and must not be interpreted as measured sidewall-scattering loss. The real group index is used in the reported propagation estimates.

D. Historical baseline and coverage

The 5 October 100-cell calculation gave $n_g = 4.245970$ and contained group-index steps near 1532.6 and 1600 nm. It remains historical evidence; its complete 1500-1600 nm vector is not spliced into the newer fitted sweep. Later MODE extension points include 1600 nm, but do not fill the missing dense 1550-1600 nm interval.

The earlier 600 × 220 nm exercise gave $n_{eff} = 2.571982$ and $n_g = 4.104976$ at 1550 nm. That separate width is retained as preliminary context and is excluded from the v11 process statistics.

XI. Geometry And Thermal Sensitivity

A. Nine geometry corners

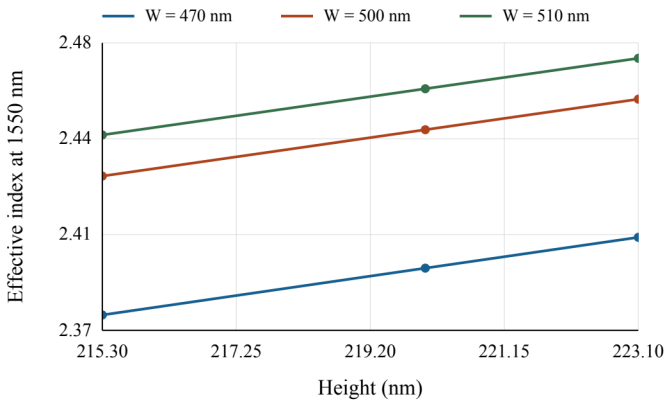


Fig. 14. Fitted 400-cell MODE results for the nine specified geometry corners at 1550 nm and nominal 300 K. Lines connect the three solved heights for readability.

The corners combine widths 470, 500 and 510 nm with heights 215.3, 220 and 223.1 nm. Effective index ranges from 2.37621823 to 2.47252875, while TE polarization fraction ranges from 97.9373% to 98.5010%. Group-index extrema occur at different corners from effective-index extrema.

One reported overlap-to-nominal value reaches 1.00026315. The original value is retained rather than clipped to unity; the existing upper-bound audit identifies a normalization inconsistency. Complete field vectors are not available to independently resolve its cause.

B. Uniform temperature estimate

The MODE temperature model assumes dn/dT = 1.9 × 10^{-4}/K for silicon and 8.5 × 10^{-6}/K for the background, referenced to 300 K. No heater geometry, thermal gradient, stress or expansion is included.

The latest 800-cell endpoints at 300 and 310 K give Δn_{eff} = 0.002047298863 and Δn_g = 0.002314041093. At fixed geometry and coupler phase, the first-order fringe estimate is

$$\Delta\lambda \approx \lambda\, \Delta n_{eff}/n_g = +0.753 \text{ nm}. \quad (12)$$

This corresponds to approximately 75.35 pm/K and a fixed-1550-nm FSR change of -0.054915%. It is an analytic propagation estimate for a uniform +10 K change, not a simulated full-circuit thermal spectrum.

C. Native thermal diagnostics

The native A25 300/310 K precheck has zero maximum complex-transmission and power differences. For both Bragg batches, all three 305 K and 310 K spectra are also pointwise identical to the 300 K nominal spectra. The native index lookup and public Bragg parameters do not change.

These diagnostics show that the current native configuration does not produce a thermo-optic response. They do not demonstrate that the physical circuits are temperature insensitive. The estimate in (12) has not been inserted into the native scattering models.

XII. Response Model Accuracy

The response69 study contains 60 geometry/temperature calibration solves, eight independent holdouts and one 305 K midpoint. Bilinear geometry interpolation and endpoint temperature interpolation generate a separate exploratory 10,000-sample surrogate ensemble.

TABLE VII
INDEPENDENT SURROGATE HOLDOUT ERRORS

Metric	Maximum error	Target
$\lvert\Delta n_{eff}\rvert$	3.68966 × 10^{-4}	10^{-5}
$\lvert\Delta n_g\rvert$	2.78903 × 10^{-4}	10^{-3}
Phase-equivalent shift	0.135728 nm	0.05 nm
Local FSR relative error	0.0066267%	0.2%

The effective-index and phase targets are not met. Consequently, the surrogate wavelength distributions remain provisional. A cubic diagnostic did not remove this limitation; higher interpolation order is not evidence of improved accuracy.

The additional extension80 dataset contains 36 bus wavelength anchors, 38 uniform-section width-extension solves and six fine-mesh checks. It extends evidence to 450-550 nm cross sections and discrete 1500/1550/1600 nm anchors. These are uniform-waveguide modes, not periodic-grating scattering solutions.

Neither extension80 nor the surrogate replaces the 100 saved native circuit simulations. MODE69 conditions its height distribution on 207.5-230.9 nm, whereas native MC100 uses 210-230 nm. Their models, sample counts and conditional domains differ. Their percentiles must not be pooled or presented as repeated estimates of one population.

XIII. Native Process Study

The 42 deterministic rows comprise one nominal case, 15 width steps toward 470 nm, five toward 510 nm, six height steps toward 215.3 nm, six toward 223.1 nm, and nine W/H corners. Width sweeps hold H = 220 nm; height sweeps hold W = 500 nm. The corner set uses W = 470/500/510 nm and H = 215.3/220/223.1 nm. The nominal corner is intentionally repeated as a separate role, so 42 counts cases rather than unique geometries.

All five native circuits use the same 100 geometry trials (seed 20261011). Width is uniform on 470-510 nm. Height follows a Normal(219.2 nm, 3.9^2 nm^2) parent distribution conditioned by rejection sampling on 210-230 nm. Width and height are sampled independently.

Each die applies common width/height offsets to its controlled waveguides; the Bragg model receives the same offsets. Couplers and grating couplers remain fixed. Period, corrugation, misalignment and random local sidewall-loss variation are not sampled. Native width-dependent propagation loss remains active.

The observed sample means are 490.946464 nm in width and 218.726995 nm in height, with sample standard deviations 11.094920 and 3.712510 nm. The width parent mean is 490 nm, below the 500 nm nominal design, so a negative average spectral shift is expected under these inputs.

A. MZI metric extraction

Power is recomputed as $|t|^2$ from the saved complex transmission. Strict local minima are refined by three-point nonuniform parabolic interpolation. Principal minima are selected separately from weak interference ripples; depth thresholds are checked for selection stability.

A25 uses a stable $10^{-6}/10^{-5}/10^{-4}$ power-threshold family. A100 uses $10^{-4}/3 \times 10^{-4}/10^{-3}$, since deeper thresholds omit principal fringes. Course500 uses local relative prominence 0.5, with consistency checks at 0.4 and 0.6.

Adjacent selected minima give frequency intervals. Interpolation between their midpoint frequencies to $f_0 = c/1550$ nm gives the local equivalent FSR

$$FSR_\lambda(\lambda_0) = \lambda_0^2 \, \Delta f(f_0)/c. \quad (13)$$

All cases bracket this interpolation point. This definition differs slightly from the earlier direct sampled-minimum estimates. Three-point refinement does not establish independent spectral-resolution convergence or the limiting depth of very sharp notches.

XIV. Nominal MZI Results

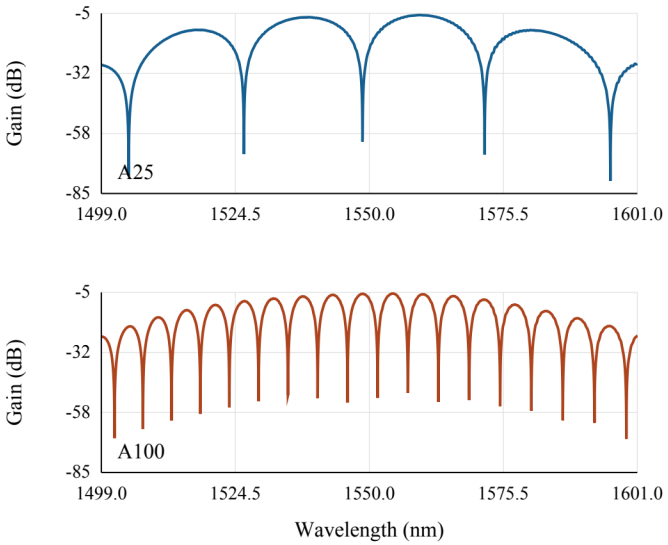


Fig. 15. Native nominal A25 and A100 external gain, each reconstructed from 5,000 complex samples. The grating-coupler envelope and secondary ripple are retained.

TABLE VIII
NOMINAL LOCAL MZI METRICS

Circuit / output	ΔL (μm)	FSR (nm)	Ref. notch (nm)
A25	25.000	22.93070	1548.71755
A100	100.000	5.73255	1551.58446
Course500 / 1	99.294	5.83856	1549.65442
Course500 / 2	99.294	5.83600	1552.57921

The A25/A100 FSR ratio is approximately four, consistent with inverse path-imbalance scaling. The new extraction yields 22.930701 and 5.732553 nm; the 5 October rounded values were 22.934 and 5.734 nm. This is an extraction-definition refinement, not a layout revision.

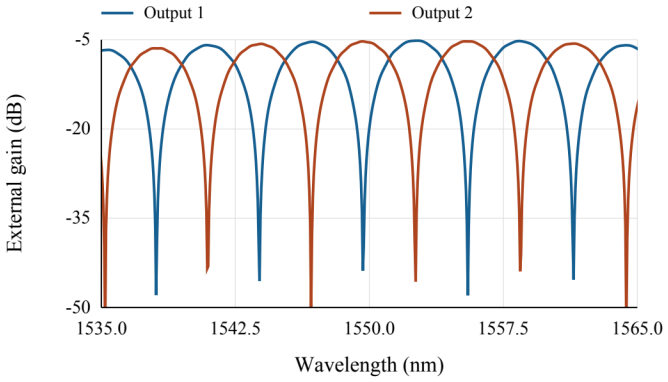


Fig. 16. Course500 nominal outputs over a 30 nm window. The two complementary fringes include the full circuit response; each complete source trace contains 2,000 points.

Using native $n_g = 4.19088$ and $\Delta L = 99.294$ μm in (3) gives 5.773447 nm, about 1.12% below Course500 output 1. Coupler differential phase and complete-circuit extraction cannot be omitted from this comparison.

XV. Paired Monte Carlo Results

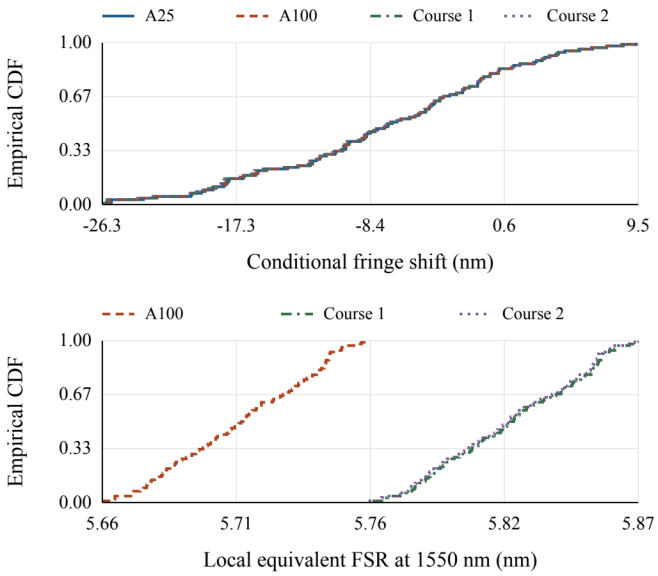


Fig. 17. Paired native MC100 empirical cumulative distributions. Corresponding-fringe shifts are conditional on model-assisted branch association; A25 FSR is tabulated separately because its scale is about four times larger.

The same trial IDs permit direct circuit comparisons. A100 and Course500 output 1 FSR have a paired correlation of 0.99953852. A25/A100 conditional fringe shifts have correlation 0.99999924. These are model results under common die offsets and fixed couplers, not measured wafer correlations.

A. Corresponding-fringe association

A native-index phase hint is combined with spectral candidates and small-step width/height continuation. Each continuation step remains below half the local FSR. Course500 additionally agrees with nearest-prior tracking along the supplied continuation paths.

Twenty-nine MC heights lie outside the dense 215.3-223.1 nm continuation interval. Their branch labels remain model assisted. Thus the reported shifts are conditional corresponding-fringe shifts; no independently verified absolute fringe order is claimed for all trials. FSR extraction does not require such an absolute order.

TABLE IX
NATIVE MC100 FSR STATISTICS (NM)

Output	Mean	SD	P2.5 / P97.5
A25	22.85164	0.10968	22.65645 / 23.02278
A100	5.71296	0.02741	5.66417 / 5.75576
Course 1	5.81881	0.02780	5.77039 / 5.86267
Course 2	5.81732	0.02781	5.76838 / 5.86125

TABLE X
CONDITIONAL FRINGE-SHIFT STATISTICS (NM)

Output	Mean	SD	P2.5 / P97.5
A25	-7.42537	8.28320	-24.66495 / 6.93826
A100	-7.46207	8.32206	-24.78128 / 6.95902
Course 1	-7.51542	8.38392	-24.95666 / 7.02775
Course 2	-7.55394	8.42595	-25.08344 / 7.05006

All entries in Tables IX-X use 100 native trials. SD is the sample standard deviation. Percentiles are empirical linear-interpolation summaries of this finite ensemble, not confidence bounds, population coverage guarantees or fabrication-yield estimates.

B. Design interpretation

A25 has a larger absolute FSR spread because its FSR is about four times that of A100. Their wavelength-shift spreads remain similar (about 8.3 nm), consistent with a common propagation-index perturbation. A comparable wavelength shift therefore spans more fringes in the shorter-FSR circuits.

Nine-corner FSR ranges are 22.61916-23.04944 nm for A25 and 5.65499-5.76230 nm for A100. Course500 ranges are 5.75993-5.86926 nm and 5.75799-5.86768 nm for outputs 1 and 2. Complete paired corner values are retained in the appendix.

The negative mean shifts follow the selected process distribution, whose width centre is below nominal. They should not be described as an unexplained simulator bias. The three MZI datasets also cannot be pooled as 300 independent trials: all share the same 100 geometry inputs.

No common acceptance windows for FSR, wavelength, insertion loss or extinction were specified. Accordingly, the report provides sensitivity distributions rather than a single yield percentage.

XVI. Bragg Design Rationale And Model Equations

A. Period, corrugation and interaction length

A periodic index perturbation couples forward and backward waves. First-order phase matching gives the familiar approximate Bragg relation

$$2\beta \approx 2\pi/\Lambda,\ \lambda_B \approx 2\Lambda n_{avg}. \tag{14}$$

The index n_{avg} represents the periodic structure in this approximation; it is not automatically the straight bus-waveguide index. The native model uses an empirical centre correction, given explicitly in the next column. That implemented expression controls the reported model-centre values.

Both filters retain $\Lambda = 317$ nm and corrugation parameter a = 50 nm. The controlled variable is N: 150 and 300 periods give lengths $L = N\Lambda$ of 47.55 and 95.10 μm. Holding period and corrugation fixed separates an interaction-length comparison from a deliberate centre-frequency redesign.

B. Why double the number of periods?

For an ideal uniform, lossless, matched grating at resonance, coupled-mode theory gives the qualitative length dependence [4]

$$R = \tanh^2(\kappa L),\ T = \operatorname{sech}^2(\kappa L). \tag{15}$$

The native nominal κ is 75,375.25 m^{-1}, so κL is approximately 3.584 for N150 and 7.168 for N300. Increasing length strengthens rejection while ideal reflectivity approaches saturation. The physical grating length doubles, introducing a footprint cost and a longer lossy interaction region.

These ideal equations explain the expected trend, not the actual external port power. The implemented WBG has 590 dB/m loss and left/right facet phases of $+\pi/4$ and $-\pi/4$; the complete circuit also contains couplers and routing. External attenuation and normalized stopband suppression must therefore be reported separately.

The simulated N300 result is deeper suppression with a narrower nominal half-power band under the stated normalization. This is the observed tradeoff for these two circuits; it is not a universal claim that doubling N always halves bandwidth or optimizes a filter specification. No rejection or bandwidth acceptance threshold was supplied.

C. Exact native centre calculation

The installed rectangular-grating setup (model version 2020/02/11) interpolates n_e and n_g from its waveguide lookup at 1550 nm for the chosen width and height. Let $\lambda_0 = 1550$ nm, $\Lambda_0 = 317$ nm and $n_{e0} = 2.44553$. The implemented calculation is

$$\lambda_{base} = \lambda_0[1 + ((\Lambda - \Lambda_0)/\Lambda_0)(n_e/n_g)]. \tag{16}$$

$$\lambda_{model} = \lambda_{base}(n_e/n_{e0}) - C(a),\quad C(a) = (190\ m^{-1})a^2 + 0.0259a + 10^{-9}\ m. \tag{17}$$

All lengths in these expressions are in metres. With longitudinal misalignment δ, the source also sets

$$\kappa = [-(1.53519 \times 10^{19}\ m^{-3})a^2 + (2.2751 \times 10^{12}\ m^{-2})a]\cos(\pi\delta/\Lambda). \tag{18}$$

At the nominal geometry, $n_e = n_{e0}$, $n_g = 4.19088$ and $\delta = 0$. Thus $\lambda_{base} = 1550$ nm, and the three correction terms are 0.000475, 1.295 and 1 nm:

$$\lambda_{model} = 1550 - 2.295475 = 1547.704525\ nm. \tag{19}$$

In contrast, directly inserting the bus index into $2\Lambda n_e$ gives 1550.46602 nm. The discrepancy is explained by the native calibration and corrugation correction; it is not resolved by silently replacing the implemented model with the simple phase-matching approximation.

D. How geometry variation enters

The saved cases change width and height, hence the lookup indices and λ_{model}. Period, corrugation and misalignment remain fixed, so κ is fixed in this native comparison. N changes L, not the centre equation. Paired N150/N300 cases therefore share the model-centre distribution even though their transmission extrema and half-power midpoints can differ.

This is a compact-model prediction with empirical coefficients. Independent periodic-grating electromagnetic calibration is absent; agreement with the source formula verifies implementation consistency, not physical accuracy of the empirical correction.

XVII. Bragg Filter Results

A. Reference-normalized suppression

Native Bragg output ports 1, 2 and 3 correspond to reflection, reference and transmission. External gain includes grating couplers, routing and directional-coupler effects. A reference ratio removes part of their common envelope:

$$Q(\lambda) = P_3(\lambda)/P_2(\lambda),\ q(\lambda) = Q(\lambda)/Q_{off}. \qquad (20)$$

Q_{off} is the median Q over available points 20-35 nm from each case's public model Bragg wavelength. The connected q < 0.5 region containing the central minimum is used. The minimum search is restricted to ±10 nm around that model wavelength; crossings are linearly interpolated in power ratio.

The half-power threshold is -3.01030 dB, abbreviated here as the 3 dB suppression band. Width is right crossing minus left crossing; the band midpoint is their average. A missing crossing is censored, not extrapolated. The public model wavelength, band midpoint and deepest sampled point are distinct quantities.

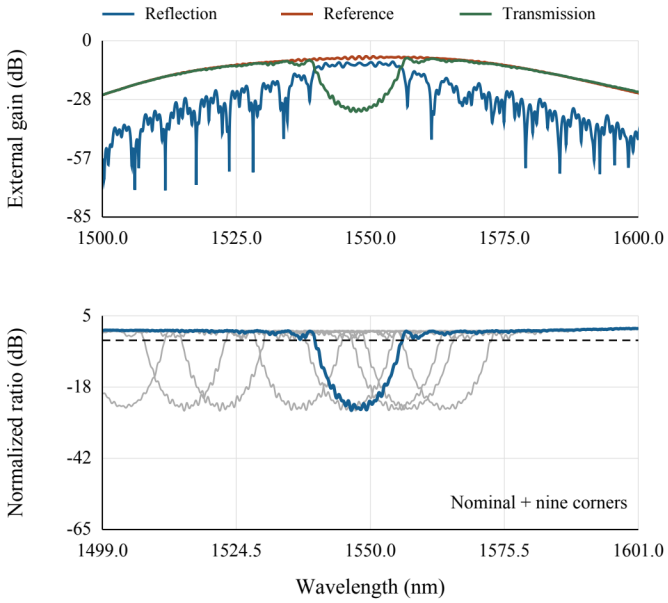


Fig. 18. N150 nominal external channels (top) and nominal/geometry-corner normalized transmission ratios (bottom). Grey traces show nine corners; the dashed line is the half-power threshold.

This normalization defines external circuit suppression relative to the reference branch. It is not independent de-embedding of intrinsic grating transmission or reflectivity.

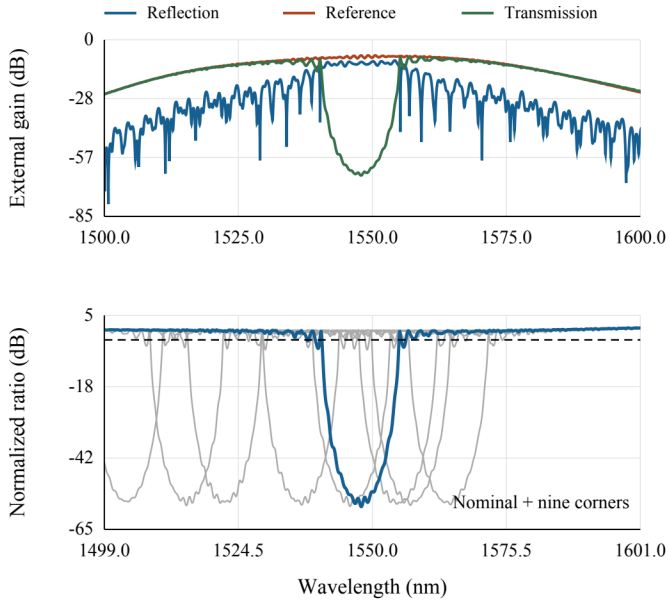


Fig. 19. N300 spectra from the valid map200 batch. The same metric convention is used as for N150; deeper suppression and a narrower nominal half-power band are observed.

TABLE XI
NOMINAL BRAGG COMPARISON

Quantity	N150	N300
Length (µm)	47.55	95.10
Model wavelength (nm)	1547.70453	1547.70453
Band midpoint (nm)	1547.77220	1547.74279
3 dB width (nm)	16.64318	14.67834
Sampled suppression (dB)	25.98745	57.56015
External gain at 1550 (dB)	-31.99668	-61.47789
Left crossing (nm)	1539.45061	1540.40362
Right crossing (nm)	1556.09379	1555.08196

At 1550 nm, N300 external transmission is 29.48 dB below N150. Under the specified relative half-power definition, its nominal band is 14.6783 nm wide, versus 16.6432 nm for N150. More periods therefore do not imply a larger width under every bandwidth convention.

The nominal public coupling coefficient is 75,375.25 m^{-1} for both devices. The native empirical model fixes period, corrugation and grating loss (590 dB/m). The uniform-section MODE extension does not independently validate this coupling coefficient or the grating rejection depth.

XVIII. Bragg Process Variation

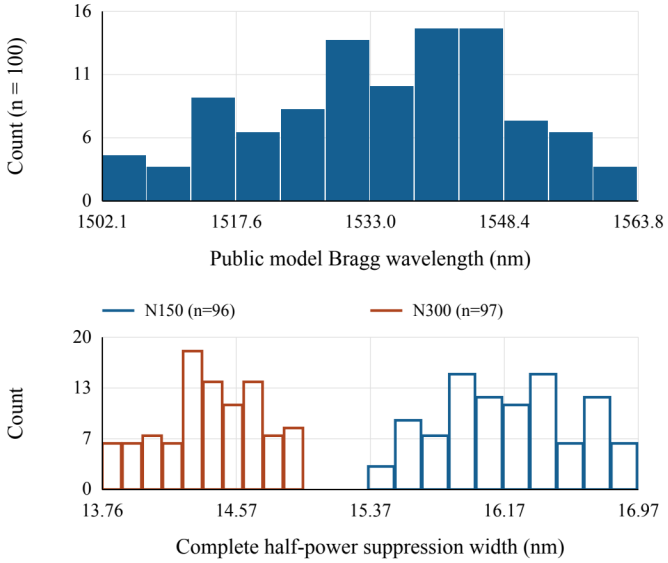


Fig. 20. Paired native MC100: the common public model wavelength uses all 100 trials. Half-power bandwidth histograms include only complete two-crossing cases: 96 for N150 and 97 for N300.

TABLE XII
BRAGG MC STATISTICS

Metric	n	Mean	SD
Model wavelength (nm)	100	1534.72558	14.48902
Model shift (nm)	100	-12.97894	14.48902
N150 width (nm)	96	16.19397	0.39880
N300 width (nm)	97	14.41972	0.30830
N150 suppression (dB)	96	25.98102	0.12929
N300 suppression (dB)	97	57.01508	0.34975

N150 and N300 have identical public model centres for paired geometries because period, corrugation and the index lookup are shared. Period count changes grating length and response depth; it does not change the public centre formula in this model.

The common model-wavelength P2.5/P50/P97.5 values are 1504.25005/1535.48574/1559.57732 nm. This parameter distribution is distinct from an independently extracted spectral-centre distribution.

A. Truncation and baseline sensitivity

The saved Bragg band is 1499.0191-1601.0000 nm. N150 trials 16, 28, 49 and 51 lack the left half-power crossing; N300 trials 16, 49 and 51 have the same limitation. One corner (470 × 215.3 nm) is also left censored for each device.

Their spectra and public model centres remain available, but missing widths are not filled using a coupling-coefficient estimate. Width and suppression-depth statistics in Table XII describe the complete-crossing subsets, not unbiased distributions for all 100 trials.

For complete widths, P2.5/P50/P97.5 are 15.51786/16.19145/16.84862 nm for N150 and 13.85679/14.44107/14.92118 nm for N300. A wider short-wavelength sweep would be needed to report every trial's full band.

TABLE XIII
NOMINAL WIDTH VERSUS OFFBAND WINDOW

Distance from model centre	N150 (nm)	N300 (nm)
15-30 nm	16.61031	14.67153
20-35 nm (adopted)	16.64318	14.67834
25-40 nm	16.67096	14.68551

Across common complete cases, changing the baseline window gives maximum width differences of 0.10830 nm for N150 and 0.00914 nm for N300. These quantify extraction-definition sensitivity only; they exclude model calibration and fabrication uncertainty.

B. N300 simulation setup verification

An earlier 100 × 100 wafer map placed the N300 grating outside the valid index range and left public model outputs empty. The adopted result is the independently rerun 200 × 200 map at the same 10 µm grid, with required parent properties restored.

The grating and five controlled waveguide map indices are within 1-200. All 144 cases have nonempty public results consistent with the native formula checks. A second-case spectrum-shift guard confirms that the result changes with geometry. Failed-setup and stale-spectrum attempts are excluded from the statistical batch.

XIX. Verification And Limitations

The deterministic evidence audits decode the MATLAB v7.3/HDF5 complex arrays, check finite values, actual trace lengths, fλ = c, shared grids, case metadata and prepared geometry inputs. FSR/minimum extraction is recomputed from full arrays. Synthetic fringe and half-power-band references test the extraction algorithms.

These checks establish data and algorithm consistency within their stated scope. They do not independently calibrate the compact models against measurements or full-wave periodic-grating solutions. The report preserves failed numerical targets rather than equating readable data with validated device physics.

TABLE XIV
EVIDENCE LIMITS RETAINED IN THIS REVISION

Topic	Current boundary
Material fit	RMS exceeds the requested tolerance; retained provisionally.
MODE mesh	Six n_{eff} refinement differences exceed 10^{-5}.
MODE interpolation	Holdout maxima exceed n_{eff}/phase targets.
Mode overlap	Some corners exceed unity; normalization unresolved.
Dense MODE band	Updated export ends at 1550 nm.
Native temperature	Zero-response diagnostics; no calibrated thermal circuit.
MZI fringe identity	29 MC heights outside dense continuation.
Bragg tails	4/3 censored MC widths for N150/N300.
Physical validation	No fabrication data or foundry signoff.

Exact fitted-material project persistence, all-geometry convergence and independent periodic-grating electromagnetic validation remain incomplete. The native terminal-landing treatment is also not separately characterized. These limits affect the strength of the physical predictions, not the identity of the saved spectra.

A future physical validation should first establish consistent material and coupler models, implement nonzero thermo-optic response in the native circuit, and extend the Bragg short-wavelength range. Yield requires explicit acceptance criteria and a justified process distribution before further sampling can be interpreted as yield evidence.

XX. Conclusion

The retained preliminary 600 nm study connects modal confinement, dispersion fitting and ideal-MZI scaling. The original 500 nm baseline documents its full sweep and unresolved anomalies. The 11 October study extends this foundation with finer-mesh MODE comparisons and complete native circuit ensembles; it does not retroactively validate the earlier models.

The selected comparisons answer distinct design questions: A25/A100 test inverse-imbalance FSR scaling; Course500 compares a different combining topology at nearly equal imbalance; N150/N300 test stronger rejection against doubled grating length. These results explain the design tradeoffs without claiming an unperformed optimization.

The v11 design now has complete saved nominal, corner and MC100 circuit evidence for three MZIs and two Bragg filters. A25/A100 preserve approximately four-to-one FSR scaling. Course500 provides two numerically characterized outputs with full-circuit FSRs near 5.84 nm, differing from the propagation-only estimate.

N300 produces 29.48 dB lower external transmission at 1550 nm than N150 and deeper reference-normalized suppression. With the adopted half-power definition, its nominal band is narrower. Paired geometry variation shifts all circuit responses substantially, with correlated MZI wavelength changes and a common Bragg model-centre distribution.

The fitted MODE study improves the documented dispersion evidence and quantifies mesh sensitivity. Its +10 K estimate of +0.753 nm is separate from native circuit diagnostics, which currently show no thermal response. The reported results support a conditional simulation study while preserving the numerical and physical coverage gaps needed for a defensible interpretation.

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APPENDIX: CORNER COMPARISONS AND NUMERICAL SENSITIVITY

All FSRs below are local equivalent values at 1550 nm. The nine corners are discrete design cases, not guaranteed process extrema. Conditional fringe labels retain the model-association limits described in Section XV. W and H are in nm.

TABLE XV
NINE-CORNER MODAL INDICES AND FULL-CIRCUIT FSR

W	H	MODE n_{eff}	MODE n_g	Native A25 FSR (nm)	Native A100 FSR (nm)	Native Course 1 FSR (nm)	Native Course 2 FSR (nm)
500	220	2.4457219	4.2112388	22.93070	5.73255	5.83856	5.83600
500	215.3	2.4283318	4.2070814	22.95968	5.73989	5.84599	5.84481
500	223.1	2.4571880	4.2145643	22.91552	5.72894	5.83558	5.83386
470	220	2.3937866	4.2657183	22.63620	5.65908	5.76488	5.76299
470	215.3	2.3762182	4.2608832	22.66857	5.66699	5.77432	5.77224
470	223.1	2.4053512	4.2695137	22.61916	5.65499	5.75993	5.75799
510	220	2.4610953	4.1944196	23.02134	5.75534	5.86284	5.86095
510	215.3	2.4437628	4.1904033	23.04944	5.76230	5.86926	5.86768
510	223.1	2.4725288	4.1976441	23.00654	5.75175	5.86022	5.85850

TABLE XVI
NINE-CORNER BRAGG HALF-POWER BANDS

W	H	Model centre (nm)	N150 width (nm)	N300 width (nm)	Band coverage
470	215.3	1503.18590	Censored	Censored	Left truncated
470	220	1514.64186	15.62104	13.84828	Both complete
470	223.1	1521.86813	15.68249	13.99464	Both complete
500	215.3	1536.35550	16.43184	14.49232	Both complete
500	220	1547.70453	16.64318	14.67834	Both complete
500	223.1	1554.85790	16.79047	14.82671	Both complete
510	215.3	1545.98741	16.67980	14.72400	Both complete
510	220	1557.30198	16.93477	14.91969	Both complete
510	223.1	1564.43318	17.07151	15.08740	Both complete

TABLE XVII
SIX-POINT MESH SENSITIVITY

W / H (nm)	T (K)	Δn_{eff} (800-400)	Δn_g (800-400)	Phase scale (nm)	Coarse FSR excess (%)
500 / 220	300	9.01801e-05	3.01730e-04	0.033190	0.007165
470 / 207.5	300	8.44212e-05	3.39448e-04	0.030798	0.007990
510 / 230.9	300	9.90436e-05	2.54467e-04	0.036524	0.006054
500 / 220	310	9.04166e-05	3.01674e-04	0.033258	0.007160
470 / 207.5	310	8.46720e-05	3.39436e-04	0.030871	0.007985
510 / 230.9	310	9.92523e-05	2.54418e-04	0.036582	0.006050

The final column is $100(FSR_{400} - FSR_{800})/FSR_{800} = 100(n_{g,800}/n_{g,400} - 1)$, at fixed wavelength and imbalance. Positive values mean the coarse mesh predicts a larger FSR. They are not the signed 800-minus-400 refinement change; that alternative convention is -p/(1 + p/100)% for tabulated p.

MODE indices and native circuit FSRs in the first table come from different model families and are juxtaposed for comparison, not substituted into one another. These numerical results are generated in the present study. The equations, case definitions and conventions needed to interpret this PDF are stated here.

The mesh phase scale is a first-order propagation conversion of a discretization difference, not an independently measured circuit-wavelength error. Some extension geometries exceed the native MC100 height domain. The displayed precision permits numerical comparison; it does not imply equivalent physical accuracy.